內波即時告警技術及系統設備開發(2/3)

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摘　　要

本計畫旨在建置南海內波即時告警能力，透過浮標式感測、衛星/電郵鏈路回傳、雲端資料庫與告警模組，形成從「感測—傳輸—判識—告警」的一體化架構。第一年完成觀測錨碇串列設計與海試，T/P 在多深度穩定捕捉等溫面起伏與深層葉綠素極大上抬/下陷特徵，證實海域層化顯著且內波活動頻仍。兩次颱風通過佈放點，驗證通訊與資料流程可靠，同時揭示主纜在強風浪與礁岩地貌條件下之脆弱點。據此提出討論：採高張力 Dyneema 主纜與防磨配置、主站南移至平坦海盆、增設兩組驗證浮標、優化 SOP，並持續強化達觀軍艦載台之可拆卸式佈放裝備與年度維保/改裝。結合國防先進科技研究計畫之產學研能量，可加速告警系統定型化，提升海軍自主維運、潛艦航安與任務遂行能力。

關鍵詞：內波偵測、即時告警系統、感測資料庫、浮標監測、海軍作業應用

Research on the development of real-time internal wave warning technology and system equipment

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ABSTRACT

This project establishes a real-time internal-wave (IW) early-warning capability in the northern South China Sea through an integrated chain of buoy-based sensing, satellite/e-mail telemetry, a cloud database, and alert modules. In Year-1, we designed and deployed a moored array with multi-level temperature/pressure (T/P) sensors that captured robust thermocline undulations and vertical displacements of the deep chlorophyll maximum, confirming strong stratification and frequent IW activity along the shelf-slope region. Two typhoons directly crossed the deployment site, stress-testing communications and the data pipeline while exposing a major vulnerability: main-line breakage driven by large wave-induced excursions and bottom ruggedness. Building on these findings, we propose four refinements: (1) enhance mooring resilience with high-tension Dyneema main lines, anti-chafing protection, swivels, and redundant release mechanisms; (2) relocate the primary station to a flatter basin and add two verification buoys to create a spatial/temporal redundancy and for estimating wave speed,

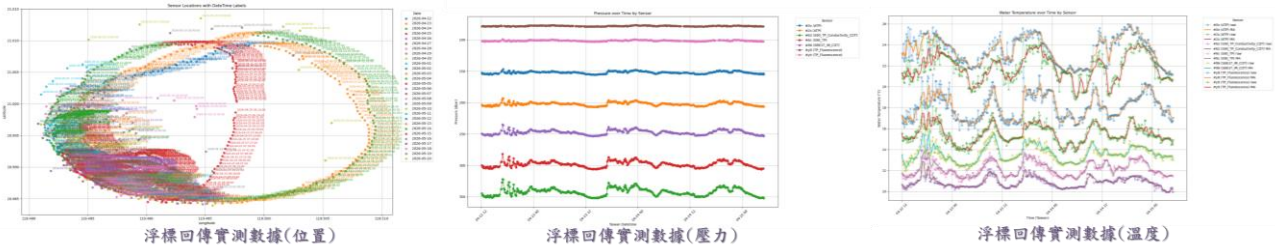
wavelength, and energy flux; (3) institutionalize ship–shore–cloud operations by equipping R.O.C. Navy R.O.C.S. Daguang with detachable deployment kits (e.g., removable bitts, double-gypsy winch, line hauler, aft derrick), accompanied by an annual maintenance/retrofit plan advised by the Ocean & Atmosphere Bureau; and (4) codify standard operating procedures, targeting higher data completeness, alert accuracy, and shorter mean-time-to-repair. Leveraging the Defense Advanced Science & Technology Program will aggregate expertise from national laboratories and industry, strengthen the Navy's in-house R&D and operations, and accelerate system hardening and scale-up. The resulting IW warning service will enhance submarine safety, operational planning, and maritime situational awareness, while advancing dual-use ocean observing capacity.

Keywords: Internal Wave Detection, Real-Time Warning System, Sensor Database, Buoy Monitoring, Naval Operational Application

The table shown below demonstrated the results of Rh-120 Barrel Performance as an example, showing how a table can be placed in the paper.



本計畫整合海洋工程、感測通訊及資料科學，分三年建構南海內波即時偵測與告警能力。於內波好發海域布建長期自動化觀測平台，回傳高頻水文與聲學資料，並由岸站進行資料處理、事件判識、視覺化及告警推播，支援海軍任務決策。計畫分為海上觀測設備開發與岸上即時告警系統建置兩大子計畫，建立「感測、傳輸、判識、告警、決策支援」之海陸一體化系統。



浮標回傳實測數據(位置)　浮標回傳實測數據(壓力)　浮標回傳實測數據(溫度)

綜整本計畫 114 及 115 年執行成果，已累積南海北部內波好發海域之溫度、鹽度、壓力（深度）等現地水文觀測資料，並結合菲律賓北方巴丹群島周邊固定潮位站長期潮汐資料，掌握潮汐變化與內波生成、傳播及抵達時間之關聯。既有資料已具備初步內波特徵及時間序列資訊，可作為告警演算法建構、參數設定及區域推估之基礎。雖 115 年海上觀測浮標因不可抗力遺失，影響長期即時資料蒐集，惟既有觀測成果仍完整保留，不影響告警系統核心研發。

116 年度將採取較具彈性及風險分散之觀測方式，依海況、航次及任務需求，運用自記式浮標或短期分批布放之深水錨碇觀測串列，持續蒐集不同季節及水深之水文資料。新增資料將與既有觀測及潮位資料整合比對，用於驗證內波抵達時間、影響範圍及強度推估，並持續修正演算法與參數，形成「實測資料—模型推估—比對驗證—系統優化」之循環機制，確保三年期計畫核心目標如期完成。